ACADEMIC CV

Hafidh Khoerul Fata

Lecturer in Mathematics · Universitas Diponegoro · Nonlinear dynamical systems and bifurcation.

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Semarang · Indonesia

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Professional profile

ROLE

Lecturer in Mathematics

AFFILIATION

Universitas Diponegoro

ACADEMIC RANK

Assistant Professor (Asisten Ahli)

PROFESSIONAL TITLE

Gr. · Mathematics

EDUCATION

2019–2021
M.Sc. Mathematics
Institut Teknologi Bandung

2018–2019
Teacher Professional Education · Mathematics
Universitas Negeri Semarang · Gr.

2013–2017
B.Sc. Mathematics
Universitas Diponegoro

RESEARCH INTERESTS

Nonlinear dynamical systems, bifurcation, delay and memory effects, mathematical modelling, and numerical dynamics.

MSC2020

- **37-XX** Dynamical systems
- **37Gxx** Bifurcation theory
- **39Axx** Difference equations
- **34Kxx** Functional-differential equations
- **37Mxx** Numerical treatment

ACADEMIC PROFILES

Google Scholar
bt3k15UAAAAJ

Scopus Author ID
59752753700

ORCID
0000-0001-6468-6118

SINTA ID
6893305

GitHub
hafidh-fata

SELECTED PROJECTS

2024–2026

2026 · **PRINCIPAL INVESTIGATOR**
Stability Analysis and Regime Change in Discrete Cyber Defense Model with Adaptive Patching

2026 · **MEMBER**
Nonlinear Dynamics and Chaos in a Banking Duopoly with Benchmark Deposit Rate Policy

2025 · **PRINCIPAL INVESTIGATOR**
Exploration of Nonlinear Dynamical Systems: Analytical and Numerical Approaches to Unveil Complex Bifurcation Phenomena

2024 · **PRINCIPAL INVESTIGATOR**
Analisis Bifurkasi Sistem Dinamik Non-Linier pada Model J.C. Sprott NE-14

Selected publications

14 JOURNAL ARTICLES · 1 PREPRINT · 1 BOOK

2026 Bifurcation structure and multistability in a sinusoidal modification of the Hénon map
Journal of Difference Equations and Applications · DOI 10.1080/10236198.2026.2689969

2026 Nonlinear Dynamics and Bifurcation Structure of a Stage-Structured Theta-Ricker Map with Cross-Stage Feedback
WSEAS Transactions on Mathematics · DOI 10.37394/23206.2026.25.3

2026 Deterministic and Stochastic Modeling of Deposit–Loan Dynamics with Optimal Regulatory Control
International Journal of Financial Studies · DOI 10.3390/ijfs14070174

2026 Threshold-Induced Bifurcation and Chaos in a One-Dimensional Coastal Risk Adaptation Map
World Journal of Advanced Research and Reviews · DOI 10.30574/wjarr.2026.30.3.1687

2025 Logistic Map-Based Banking Loan Dynamics with Central Bank Policies
Journal of Applied Nonlinear Dynamics · DOI 10.5890/JAND.2025.09.006

2025 A Study of Discrete-time Dynamical Systems with One Stable Equilibrium Using Standard and Nonstandard Difference Schemes
WSEAS Transactions on Mathematics · DOI 10.37394/23206.2025.24.64

Full publication list ↗

TEACHING

UNDERGRADUATE

Real Analysis
Limits, continuity, sequences and series of functions, and integration.

Partial Differential Equations
Classification, diffusion, waves, and analytical solution methods.

Algorithms & Programming
Problem decomposition, pseudocode, flowcharts, and Python fundamentals.

Engineering Mathematics
Linear algebra and mathematical methods for engineering problems.

CREDENTIALS

AWS Academy Graduate – Generative AI Foundations

Training badge

MCE · Microsoft Certified Educator

Gemini Certified Faculty · Google for Education

COMMUNITY SERVICE

8 ACTIVITIES · 2023–2025

Eight completed community-service activities, seven led as project chair, spanning scientific mentoring, mathematics olympiad preparation, data visualization, and local development.

- MYRES research mentoring for MAN Kendal students · 2025
- Mathematics olympiad problem-solving preparation · 2025
- Gunung Pegat tourism revitalization masterplan · 2024
- Stunting-data visualization and demographic mapping · 2023

Books & intellectual property

2023–2025

BOOK · 2025

Teori Himpunan dan Relasi

UNDIP Press · ISBN 9786324176148 · 116 pages

IP · 2023

Banking Loan Dynamics with Dividend Payments

IP · 2023

Program Diagram Eksponen Lyapunov Dinamika Kredit Berdasarkan Perubahan Deviden

IP · 2023

Program Diagram Bifurkasi Dinamika Kredit Berdasarkan Perubahan Deviden